



Integrative modeling of longitudinal cell-free DNA and tumor volume dynamics: a multimodal quantitative prognostic framework

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Background: Liquid biopsy based on cell-free DNA (cfDNA) in oncology has emerged as a promising technique for tracking cancer dynamics, especially for detecting minimal residual disease. To date, most studies have used cfDNA for static evaluations of tumor burden. In this study, we propose a novel approach integrating serial cfDNA and computed tomography (CT) tumor volume to fully reflect the dynamic nature of tumor response after treatment.

Methods: This prospective study involved 25 patients treated with curative-intent radiotherapy for localized non-small cell lung cancer (NSCLC) between June 2019 and November 2020, with 17 subsequently included in final analysis. Longitudinal blood samples were divided into two phases relative to day 3 after treatment initiation, and kinetic parameters, such as velocity and acceleration of cfDNA levels, were calculated. To complement sparse samplings in later days, volume data from routine CT scans were incorporated. K-means clustering using two different variable sets (cfDNA only and cfDNA with volume parameters) and conventional assessment using Response Evaluation Criteria in Solid Tumors (RECIST) v1.1 were applied to stratify patients, and their performance was compared.

Results: The model incorporating both cfDNA and volume parameters effectively separated responders (mean progression-free survival, 44.2 months) from non-responders [16.6 months, $P=0.02$; area under the receiver operating characteristic curve (AUC) =0.955], outperforming cfDNA only model (36.0 *vs.* 14.5 months, $P=0.04$; AUC =0.848). In contrast, RECIST v1.1-based conventional assessment showed no significant difference ($P=0.62$, AUC =0.70).

Conclusions: Therefore, our study demonstrates that integration of longitudinal cfDNA and tumor volume dynamics yielded improved assessment of treatment response and prognosis in NSCLC.

Keywords: Cell-free nucleic acid; liquid biopsy; biomarker, tumor; longitudinal study; kinetics

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Introduction

Liquid biopsy based on cell-free DNA (cfDNA) in oncology has been recognized as a promising technique for tracking cancer dynamics (1-3). This approach offers several advantages such as its non-invasiveness, the ability to capture integrated information from multiple tumor sites, and the feasibility of longitudinal monitoring (4,5). Given these advantages, cfDNA analysis has found applications in early detection, therapeutic-response evaluation, and disease-progression identification, making it a cornerstone for modern personalized medicine (6-8).

Among its clinical applications, the identification of minimal residual disease (MRD), defined as the presence of micro-metastases remaining after definitive treatment, using cfDNA analysis has been most widely studied (9-12). Detection and characterization of MRD through cfDNA analysis helps to identify molecular relapses prior to clinical or radiographic signs (13). This opportunity allows prompt intervention and improves patient outcomes. To further increase the sensitivity, a limited number of studies have attempted to temporarily amplify cfDNA release through treatment-induced cell death (14-16).

In contrast to recent advances in spatial targeting, a temporal approach to cfDNA analysis has yet to be established. Most current cfDNA applications are primarily restricted to detecting the presence of cfDNA before or after treatment (17-19). While this method provides valuable

insights to support its clinical potential, it only provides a static snapshot of tumor dynamics and overlooks the full spectrum of tumor response over time. To fully comprehend the role of cfDNA in oncology, a more refined, temporal evaluation of cfDNA dynamics, using longitudinal liquid biopsy samples, is essential.

To tackle this, we introduce a novel approach that considers velocity and acceleration of radiotherapy (RT)-induced cfDNA levels from the early (pre-day 3) to late phase (post-day 3) after treatment initiation to capture their dynamic changes in tumor response. In particular, cfDNA samples in our study were mostly collected within first 3 days to capture early treatment response in detail. With wider intervals between samplings in later periods due to practical limitations, we further refined our model based on our hypothesis that regularly acquired computed tomography (CT) scans could provide complementary information to missing cfDNA data on later days. To evaluate its performance, we have compared our combined model against the Response Evaluation Criteria in Solid Tumors (RECIST)-based model and cfDNA-only model. Finally, to assess the potential of model-selected variables, we compare them with conventional clinical features across the identified subgroups. We present this article in accordance with the TRIPOD reporting checklist (available at <https://tlcr.amegroups.com/article/view/10.21037/tlcr-2025-940/rc>).

Methods

Study population

This prospective, single-center study included 25 patients treated with curative-intent RT for localized non-small cell lung cancer (NSCLC) at Samsung Medical Center between June 2019 and November 2020. A subset of these subjects were reported in a previous study investigating targeted cfDNA amplification of NSCLC patients undergoing definitive RT (15). All participants participated in serial peripheral blood sampling for cfDNA analysis before and after RT. Eligible patients were included in this study if they (I) had either histologically confirmed World Health Organization-defined NSCLC with American Joint Committee on Cancer v9 stage I-IIIa disease or a clinical diagnosis of lung cancer without pathological confirmation due to diagnostic procedural risk; (II) received curative-intent RT; and (III) had serial cfDNA samples and chest CT scans collected before, during, and after RT. Patients

Highlight box

Key findings

- Integration of longitudinal cell-free DNA (cfDNA) kinetics with computed tomography (CT) volumetric parameters significantly improves treatment response stratification and prognosis prediction in localized non-small cell lung cancer (NSCLC) after curative-intent radiotherapy (RT).

What is known and what is new?

- cfDNA is a non-invasive biomarker widely used for minimal residual disease detection and prognostication.
- This study introduces a kinetic, two-phase cfDNA analysis (pre-day 3, post-day 3) and complements sparse post-day 3 cfDNA data with CT volumetric features.

What is the implication, and what should change now?

- Integration of pre-day 3 cfDNA kinetics with longitudinal CT volume metrics could serve as a more precise alternative or complement to Response Evaluation Criteria in Solid Tumors for evaluating RT response in localized NSCLC.

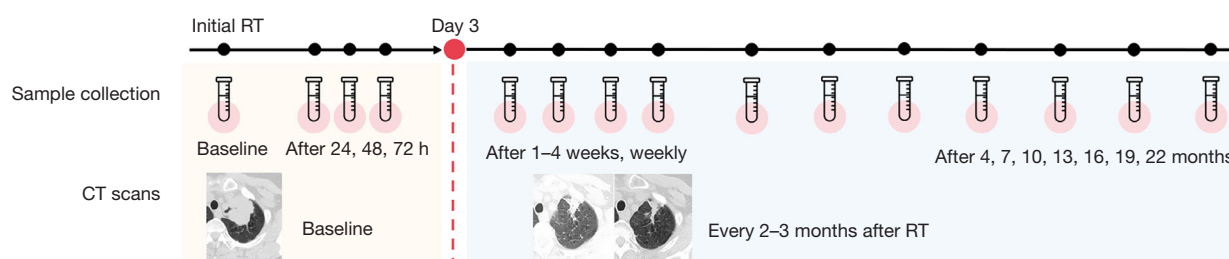


Figure 1 Conceptual demonstration of the collection method. Serial blood samples and chest CT scans were collected before and after initial RT for each patient. CT, computed tomography; RT, radiotherapy.

were excluded if they had (I) incomplete clinical data, (II) unavailable or insufficient cfDNA samples for analysis, or (III) insufficient post-RT CT images for evaluation. Demographic and clinical data, such as age, sex, smoking history, body mass index, histological type, tumor size, tumor (T) stage, node (N) stage, and radiation doses, were retrieved from the medical records.

The study was conducted in accordance with the Declaration of Helsinki and its subsequent amendments. This prospective study was approved by institutional review board of Samsung Medical Center (Nos. 2017-09-120, 2018-05155). Informed consent was taken from all the patients.

Sample preparation and cfDNA extraction

Whole-blood samples (10–20 mL) were collected in cfDNA BCT tubes (Streck, La Vista, NE, USA). Plasma was prepared through the following three centrifugation steps at room temperature with increasing centrifugal force: 840 ×g for 10 min, 1,040 ×g for 10 min, and 5,000 ×g for 10 min. All collected plasma samples were stored at 70 °C until use. Circulating cfDNA was extracted from plasma using the QIAamp Circulating Nucleic Acid Kit (Qiagen, Hilden, Germany). For validation of variants in plasma samples, LiquidSCAN (GENINUS, Seoul, South Korea) was used, targeting cancer-related genes such as *EGFR*, *TP53*, and *PIK3CA*, among others. The quantitative levels of cfDNA for all targeted genes were summed up and measured as genome-equivalents per mL of plasma. This is calculated as the product of mean variant allele frequency and the plasma cfDNA concentration in a subject divided by 0.0033 ng (20,21).

Longitudinal cfDNA sampling

Longitudinal peripheral blood samples for cfDNA analysis were collected prospectively at multiple times, including before RT (baseline); at 24, 48, and 72 h after baseline; weekly during the first month (weeks 1–4); and during post-treatment follow-up months 1, 4, 7, 10, 13, 16, 19, and 22. A total of 214 plasma samples from 25 patients were analyzed. *Figure 1* provides a conceptual demonstration of the collection method.

CT imaging acquisition and volumetric feature extraction

Chest CT imaging was performed at pre-treatment baseline and every 2–3 months after completion of RT as part of routine surveillance. All CT images were obtained with the following parameters: detector collimation, 1.25 or 0.625 mm; scans performed at 120 kVp with 150–200 mA; and reconstruction interval, 1–2.5 mm.

A region of interest (ROI) was obtained using commercial deep learning software (AviView, version 1.0.23, 2018; Coreline Soft, Seoul, South Korea) for each target lesion and corrected manually when the lesion delineation was incorrect. An additional correction was performed to exclude ground-glass opacities or bronchovascular bundles. Each ROI was confirmed by an experienced thoracic radiologist (H.Y.L.). From the ROIs, 14 quantitative CT volumetric features related to size and volume were automatically extracted. Extraction processes were conducted using the PyRadiomics package implemented in Python (<https://pyradiomics.readthedocs.io/en/latest/>) (22).

Parametrization of cfDNA dynamics

To describe the longitudinal dynamics of cfDNA over

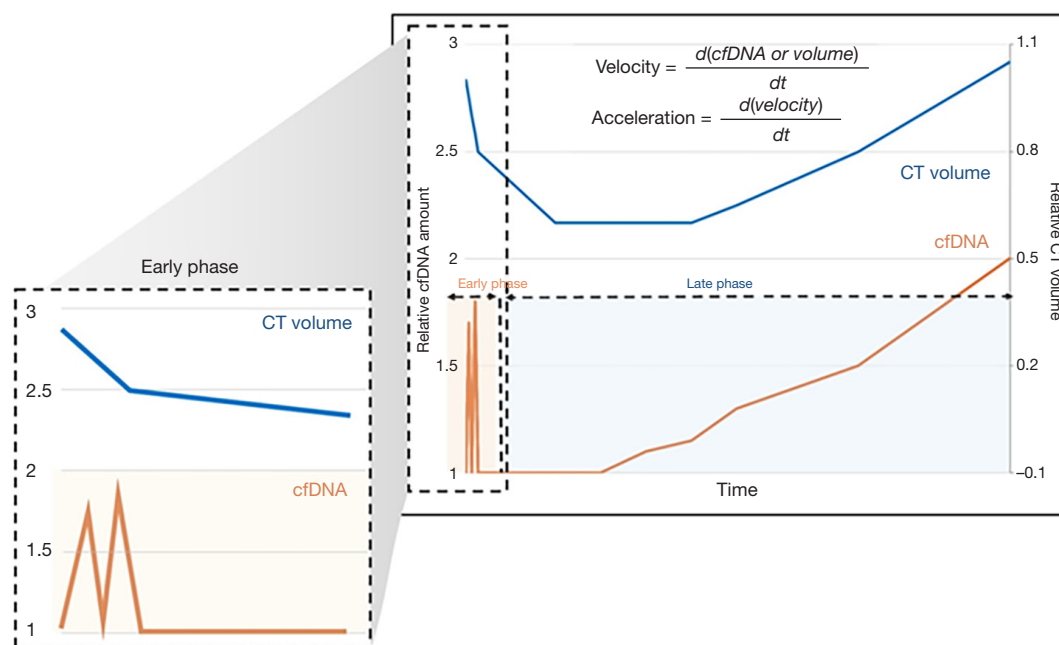


Figure 2 Conceptual graph illustrating longitudinal cfDNA and CT volume after RT. Kinetic parameters, such as velocity and acceleration, were introduced to analyze the temporal dynamics of cfDNA and CT volume. Note that the distinction between early and late phases is defined by day 3. cfDNA, cell-free DNA; CT, computed tomography; RT, radiotherapy.

time, we obtained several temporal features that capture both magnitude- and trend-related characteristics. The entire observation period was divided into two phases—an early phase (baseline to day 3) and a late phase (after day 3)—to reflect early and later dynamic changes in cfDNA. All longitudinal parameters of cfDNA were calculated separately for each period, and only patients with at least three observations both within and beyond day 3 were included. *Figure 2* presents a schematic representation of this approach.

First, we calculated basic summary statistics, including the average, minimum, and maximum values throughout the study period. To quantify the overall burden, we calculated the area under the curve with respect to the x-axis divided by days to obtain an average area per day (Area). Additionally, we measured a signed area, reflecting whether trends mostly increased or decreased over time, which was also normalized per day (auc). To depict fluctuation patterns, we defined the vertex feature, defined as the frequency of increases and decreases over time, scaled between 0 and 1. To evaluate dynamic trajectories, we computed the average rate of change (velocity) and average rate of change in velocity (acceleration), both per day.

Table S1 summarizes the definitions of all parameters.

Parametrization of CT volume

CT scans were obtained every 2–3 months after treatment. Their longitudinal dynamics of volume over time were described using the same temporal parameters as those for cfDNA. The volumetric parameters were obtained across the entire period, so at least three observations within the entire period were required for calculating the dynamic parameters.

Statistical analysis

Based on these longitudinal parameters, we applied K-means clustering to patients, with a focus on their association with progression-free survival (PFS). Clustering was performed to divide patients into two groups using (I) cfDNA-related parameters and (II) a combination of cfDNA- and volume-related parameters. In both clustering cases, patients with higher PFS values were classified as responders. For comparison, we performed clustering based on RECIST v1.1, using the longest tumor diameters at baseline and

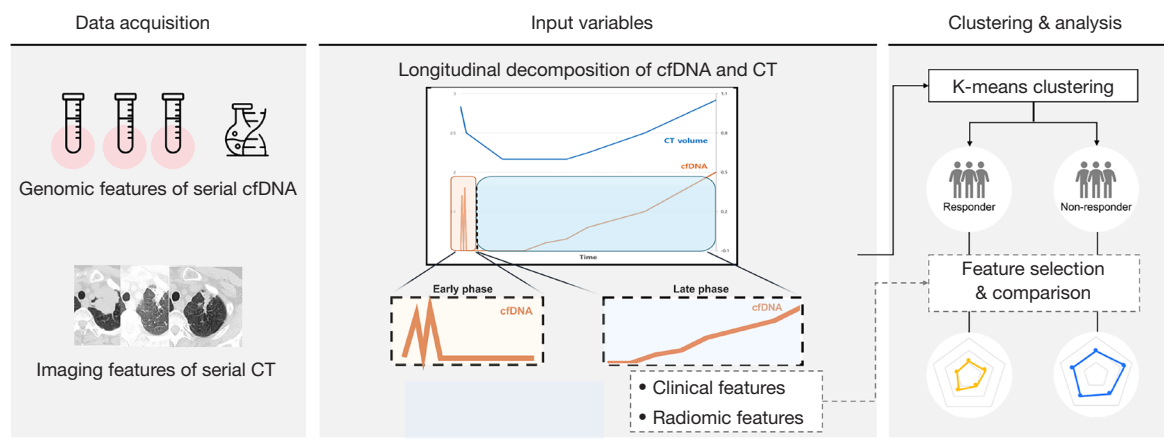


Figure 3 Overall research flow. Clinical data, serial cfDNA data, and longitudinal CT volumetric features were acquired. Longitudinal features of cfDNA and CT volume were incorporated for K-means clustering, with patients divided into two groups. Clinical features, cfDNA amount, and mode-selected variables of these two groups were compared. cfDNA, cell-free DNA; CT, computed tomography.

the first follow-up CT imaging session, recorded around 2 months after treatment, as input variables. Patients with complete or partial responses were classified as responders, while those with progressive or stable disease were classified as non-responders. Kaplan-Meier curves were drawn, and log-rank tests were conducted to compare PFS between the two groups in each clustering. For each clustering, the result with the lowest P value was selected.

For three clustering models, clinical and volumetric variables with $P < 0.1$ were selected, and multivariate logistic regression was conducted. To assess model performance, the area under the receiver operating characteristic curve (AUC) was calculated. For the model integrating cfDNA- and volume-related parameters, clinical and CT volumetric features from initial CT scans were compared using Fisher's exact test for categorical variables and the non-parametric Mann-Whitney U test for continuous variables. The overall research flow is shown in *Figure 3*.

All statistical analyses were performed using R software (version 4.4.2; R Foundation for Statistical Computing, Vienna, Austria). All reported P values are two-sided, and $P < 0.05$ was considered statistically significant.

Results

Among the initially enrolled 25 patients, 1 patient was lost to follow up, 1 patient had no follow-up CT images, and 6 patients had insufficient cfDNA samples for

analysis. As a result, a total of 17 patients were included in the final analysis.

Table 1 summarizes the three groups formed by K-means clustering with different sets of input variables. For clustering based on cfDNA parameters from early and late phases, the responder group exhibited a higher mean PFS of 36 months compared to that of 14.5 months in the non-responder group ($P = 0.04$, AUC = 0.848). Clustering based on cfDNA and volume parameters showed the greatest significance in separating patients. The responder group had a mean PFS value of 44.2 months, while the non-responder group had a mean PFS of 16.6 months ($P = 0.02$, AUC = 0.955). The Kaplan-Meier curve for this comparison is shown in *Figure 4*. Clustering based on RECIST v1.1 showed a non-significant difference, with a mean PFS of 25 months for responder and 23 months for non-responders ($P = 0.62$).

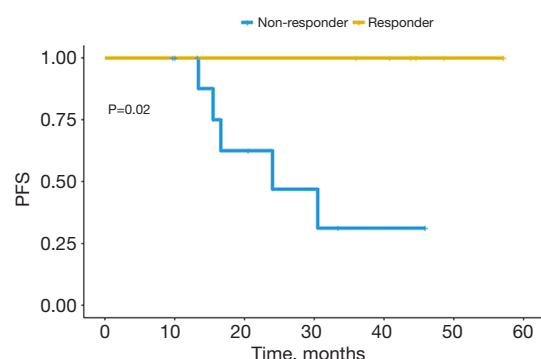
The combination of cfDNA and volume parameters—specifically, two cfDNA parameters in the early phase (auc and acceleration) and two volume parameters (vertex and velocity)—was further analyzed. *Table 2* summarizes the baseline characteristics of the responders and non-responders based on this clustering approach. The two groups did not differ significantly in age, sex, smoking status, body mass index, histological type, T-stage, lymph node metastasis, daily radiation dose, or total radiation dose. There was a borderline significant difference in tumor size ($P = 0.05$).

Figure 5 illustrates a visual comparison of selected

Table 1 Comparison of PFS across groups formed by K-means clustering with different input variables

Input variables	Group	PFS (mean), months	P value	AUC
Longest diameter (RECIST v1.1) [†]	Responder	25.0	0.62	0.70
	Non-responder	23.0		
cfDNA (t<3 & t>3) [‡]	Responder	36.0	0.04	0.848
	Non-responder	14.5		
cfDNA (t<3 & t>3) + CT volume	Responder	44.2	0.02	0.955
	Non-responder	16.6		

[†], the longest axial diameter of the target tumor was measured at baseline and first follow-up CT imaging session after RT. Patients with a complete or partial response were classified as responders, while those with progressive or stable disease were classified as non-responders; [‡], cfDNA kinetic parameters from pre-day 3 (t<3) and post-day 3 (t>3) were both included as input variables. The group with higher PFS values were designated as responders. AUC, area under the receiver operating characteristic curve; cfDNA, cell-free DNA; CT, computed tomography; PFS, progression-free survival; RECIST, Response Evaluation Criteria in Solid Tumors; RT, radiotherapy.

**Figure 4** Kaplan-Meier survival curve comparing two groups, defined by K-means clustering. PFS, progression-free survival.

clinical features, cfDNA amount, and four model-selected variables between responders and non-responders. Average cfDNA amount during the observation period, denoted as cfDNA amount, showed no statistical significance. Among four variables in our model, auc and acceleration were statistically significant ($P < 0.05$), while vertex and velocity were not. A detailed comparison of these features is summarized in Table S2.

Discussion

Previous studies have predominantly focused on the clinical utility of cfDNA in identifying MRD, based on the detectability of cfDNA before or after treatment. Bossé *et al.* (23) assessed the prognostic significance of preoperative cfDNA detection in patients with resectable

stage I NSCLC. Pan *et al.* (24) measured cfDNA levels of patients with locally advanced NSCLC undergoing chemoradiotherapy reductions as treatment continued and found that undetectable cfDNA levels on-RT and after-RT time points were associated with better prognosis. However, their method of binary detection capture provided only a static evaluation of tumor response and may not fully capture its dynamic nature.

This study explored whether longitudinal cfDNA kinetics can provide insights beyond the currently known role of cfDNA in MRD detection, by introducing a novel approach. First, we divided our longitudinal cfDNA dataset into two timeframes to reflect both early and late changes in cfDNA kinetics after RT. By collecting more samples within first three days than previous studies, we were able to capture early treatment response that may have been missed (25,26). This idea was motivated by previous studies that short-term fluctuations in ctDNA levels following the start of RT are related to early tumor cell death, while medium- and long-term ctDNA trends reflect overall tumor burden (24,27,28). Second, we calculated kinetics-related parameters such as velocity and acceleration to capture temporal shifts in cfDNA levels and obtain a more comprehensive evaluation of tumor burden changes.

Due to practical limitations, most cfDNA samples in our study were collected in an early phase, with wider intervals of collection in a later phase. To overcome the potential effects of this discrepancy, we hypothesized that incorporation of CT volume data could bridge the discontinuity of cfDNA sampling intervals in later days and enhance the overall accuracy of treatment response

Table 2 Baseline characteristics of responders and non-responders based on cfDNA and volume clustering

Characteristics	Responder (n=6)	Non-responder (n=11)	P value
Age, years	74 [72.25–75]	71 [65.5–76]	0.34
Sex			0.51
Male	6 [100]	9 [82]	
Female	0 [0]	2 [18]	
Smoking status			0.68
Current	6 [100]	11 [84]	
Former	0 [0]	2 [15]	
Never	0 [0]	1 [10]	
BMI, kg/m ²	20.7 [17.4–22.7]	22.6 [21.0–24.2]	0.31
Histological type			0.66
Squamous cell carcinoma	2 [33]	6 [55]	
Adenocarcinoma	2 [33]	1 [9]	
Missing	2 [33]	4 [36]	
Size, cm	2.3 [1.72–3.03]	3.7 [2.55–4.7]	0.05
T stage			0.89
T1	4 [66]	5 [45]	
T2	1 [17]	3 [27]	
T3	1 [17]	1 [10]	
T4	0 [0]	2 [18]	
Lymph node metastasis			>0.99
Negative	6 [100]	10 [90]	
Positive	0 [0]	1 [10]	
Daily dose, Gy	800 [800–800]	800 [400–800]	0.27
Total dose, Gy	6,400 [6,100–6,400]	6,400 [6,000–6,400]	0.96

Data are presented as median [interquartile range] values or counts [percentages]. Size is defined as the longest diameter of tumor. BMI, body mass index; cfDNA, cell-free DNA; T, tumor.

evaluation. The validity of this approach is supported by findings from Dawood *et al.* (29), who compared different diagnostic modalities and found that cfDNA had high specificity for detecting recurrence, while imaging assessments such as CT have demonstrated high sensitivity.

Our findings suggest that longitudinal cfDNA kinetics better differentiate patients between responders and non-responders than conventional RECIST-based evaluation. Furthermore, the incorporation of CT volumetric parameters along with cfDNA kinetics further enhanced model performance, implying better stratification of

patients by treatment response. This model included cfDNA parameters from the early phase and volumetric parameters from the late phase. Assaf *et al.* (30) previously showed both that early cfDNA fluctuation provides strong prognostic information and that machine learning models using cfDNA trends are superior to radiographic assessment. In accordance with these findings, our model suggests that short-term cfDNA spikes capture the early treatment response, while CT volume better suggests the tumor burden in the long term. In other words, CT imaging may serve as a complementary tool to cfDNA

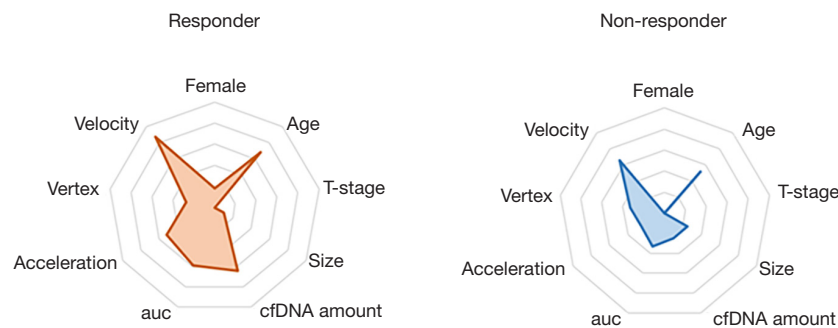


Figure 5 Radial graphs comparing clinical features, cfDNA amount, and model-selected variables between responder and non-responder group. All numerical variables are min-max normalized. Sex is shown as the percentage of female patients. 'cfDNA amount' represents average amount of detected cfDNA during the observation period. 'auc' and 'acceleration' are calculated from early phase of cfDNA kinetics, while 'vertex' and 'velocity' from volumetric kinetics. auc, a signed area reflecting whether trends mostly increased or decreased over time, normalized by day; cfDNA, cell-free DNA; T, tumor.

analysis during a late phase and integrating both modalities at different timepoints can depict tumor dynamics more precisely.

Our model uncovered four potential markers that showed discriminative potential between responders and non-responders. While conventional features such as sex, age, and T stage, size, and cfDNA amount were not statistically different between two groups, auc and acceleration, two of the selected markers, did. Although vertex and velocity were not statistically significant, they were identified as important predictors in our model, implying their potential in treatment response prediction.

Despite its strengths, our study also has several limitations. First, it was a single-center study with a small number of patients, which limits the generalizability of its results. Further validation is required in larger populations; however, the consistency with previous studies is promising. Second, our model does not take into account the tumor heterogeneity; for example, patients with KRAS-mutant NSCLC receiving KRAS^{G12C} inhibitors, compared to those with EGFR- or ALK-mutant forms, have much lower treatment response rates (31-33). Genetic variability should be considered in future work.

Conclusions

In conclusion, our study demonstrated that longitudinal cfDNA kinetics, especially when combined with CT volume, has potential as an alternative to conventional imaging-based assessment for monitoring treatment response and predicting prognosis in NSCLC. In particular,

the integration of kinetic parameters provides a deeper understanding of tumor dynamics and its response. Although further validation in larger populations is needed, this approach holds promise for potentially complementing or hopefully surpassing current evaluation methods.

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None.

Footnote

Reporting Checklist: The authors have completed the TRIPOD reporting checklist. Available at <https://tclr.amegroups.com/article/view/10.21037/tclr-2025-940/rc>

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Conflicts of Interest: All authors have completed the ICMJE uniform disclosure form (available at <https://tldr.amegroups.com/article/view/10.21037/tldr-2025-940/coif>). The authors have no conflicts of interest to declare.

Ethical Statement: The authors are accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. The study was conducted in accordance with the Declaration of Helsinki and its subsequent amendments. This prospective study was approved by institutional review board of Samsung Medical Center (Nos. 2017-09-120 and 2018-05155). Informed consent was taken from all the patients.

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